

SMART PARKING SYSTEM

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Title: Smart Parking System Using IoT and Sensors to Mitigate Traffic Issues

Objective

The primary objective of this project is to design and develop an intelligent parking system that guides drivers to available parking spots in real-time, reducing traffic congestion and improving urban mobility.

Tools Used

1. Hardware:

- Sensors: Ultrasonic or infrared sensors to detect parking spot occupancy.
- Microcontroller: Arduino or Raspberry Pi for processing sensor data.
- Wi-Fi Module: ESP8266 or similar for sending data to the cloud.
- Power Supply: Batteries or solar panels for sustainable operation.

2. Software:

- Cloud Platform: Firebase, AWS, or Google Cloud for data storage and processing.
- Mobile App: Developed using Flutter or React Native for user interface.
- Data Analytics: Tools like Google Maps API for navigation and parking spot analysis.

Methodology:

1. Parking Spot Detection:

- Install sensors in parking spots to detect occupancy.
- Sensors send real-time data to the microcontroller.

2. Data Processing and Transmission:

- Microcontroller processes sensor data and sends it to the cloud platform.
- Cloud platform analyzes data and updates parking spot availability.

3. User Interface:

- Mobile app displays real-time parking spot availability and navigation.
- Users can reserve parking spots and receive directions.

Output:

1. Real-Time Parking Availability: Users can view available parking spots on a map.
2. Navigation: Users receive turn-by-turn directions to the reserved parking spot.
3. Reduced Traffic Congestion: By guiding drivers to available spots, traffic congestion is minimized.

Result:

1. Improved Parking Efficiency: Users spend less time searching for parking spots.
2. Reduced Traffic Congestion: Fewer vehicles circling blocks in search of parking reduces traffic congestion.

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3. Enhanced User Experience: Users can plan their parking in advance and receive notifications.

Conclusion:

The Smart Parking System using IoT and sensors offers a practical solution to reduce traffic congestion and improve urban mobility. By providing real-time parking availability and navigation, this system enhances the overall parking experience and contributes to a more efficient urban environment.

Project:

<https://lovable.dev/projects/0b6387b5-4207-403b-810c-0a176cc78547>

GitHubprofile:

<https://github.com/Hemamalini49/Smart-Parking-System>